Cycle rule: trough to trough + 75 microvolts

The same rule is applied to GRDA, MASS N3 and LENS N3.

Review question

Do the retained cycles capture the clearly formed waves you want to compare?

1. Keep the existing troughs from the 0.5-4 Hz detector.

2. Between each pair, measure maximum minus minimum voltage in the 0.5-30 Hz EEG.

3. Retain cycles with peak-to-peak amplitude >=75 microvolts and duration 0.25-2 s.

Constant waves and incomplete edge fragments remain excluded.

Each example

EEG with numbered cycles above; their amplitudes and the 75-microvolt threshold below.

Teal = retained. Orange = excluded. Excluded intervals keep their original boundaries.

One familiar example, then 12 fresh examples (four per dataset; Python random seed 20260924).

Common-average reference and identical filters across datasets; this tests cycle inclusion only.

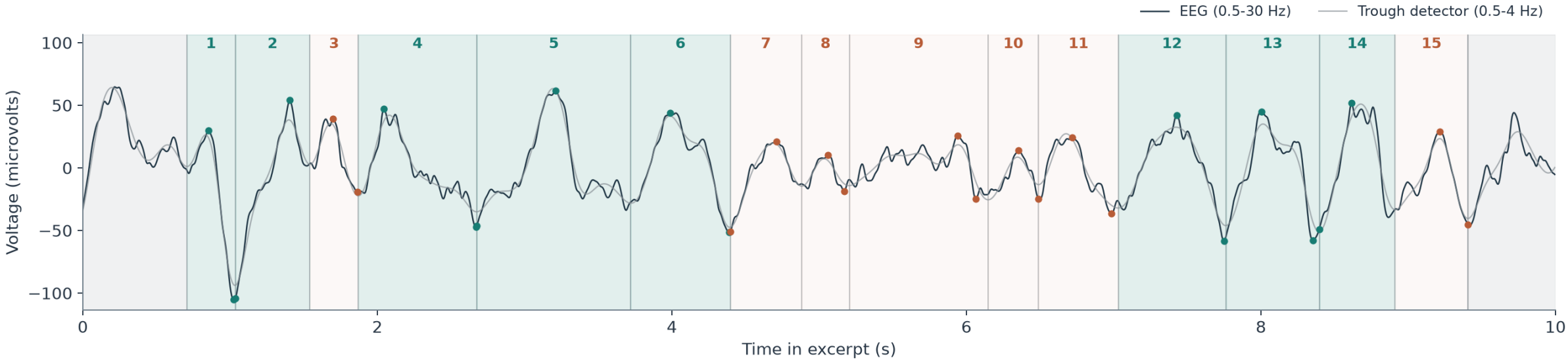
Units: MASS/LENS use EDF calibration. NESI assumes the published export values are microvolts.

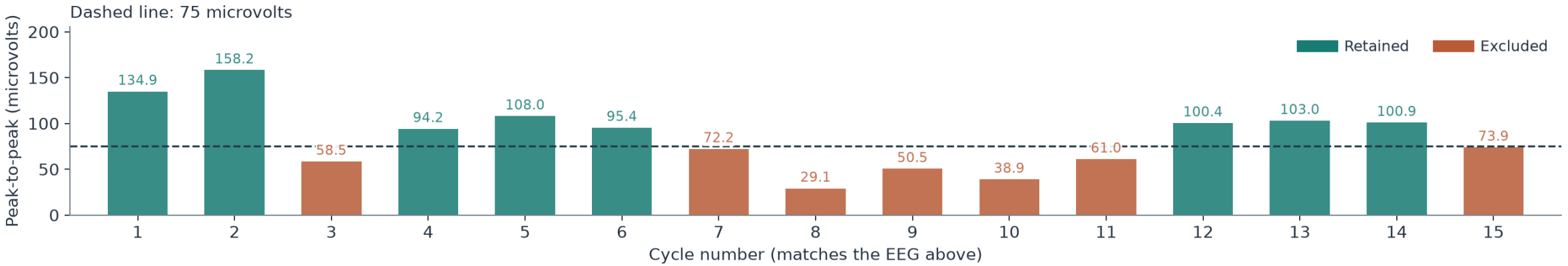
The main comparison measurements are preserved. Small differences around 75 use unrounded values.

Which cycles does 75 microvolts retain?

MASS-REFERENCE | MASS N3 | Fp1-CAR | Your previously discussed example (old cycle PDF page 7)

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





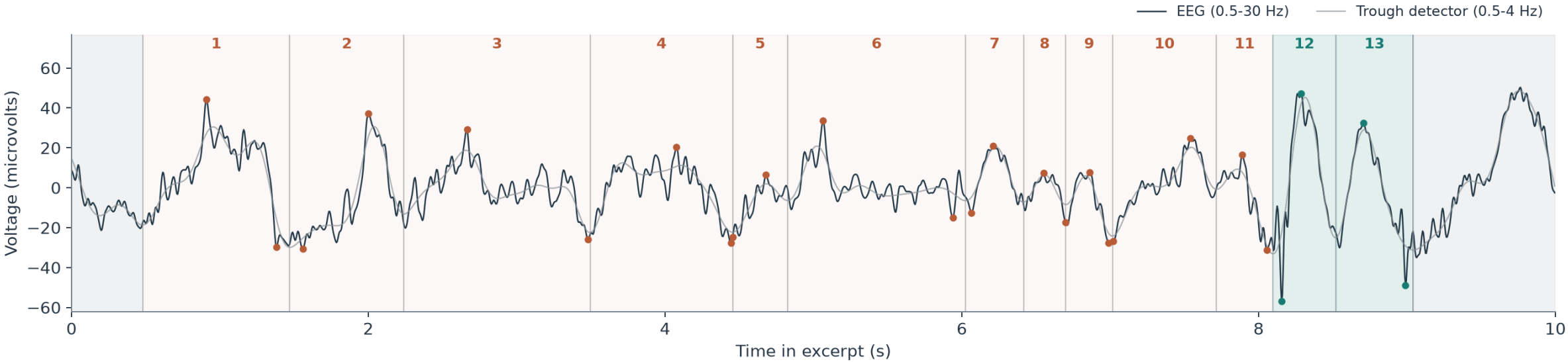
Retained: 8 of 15 previously accepted cycles.

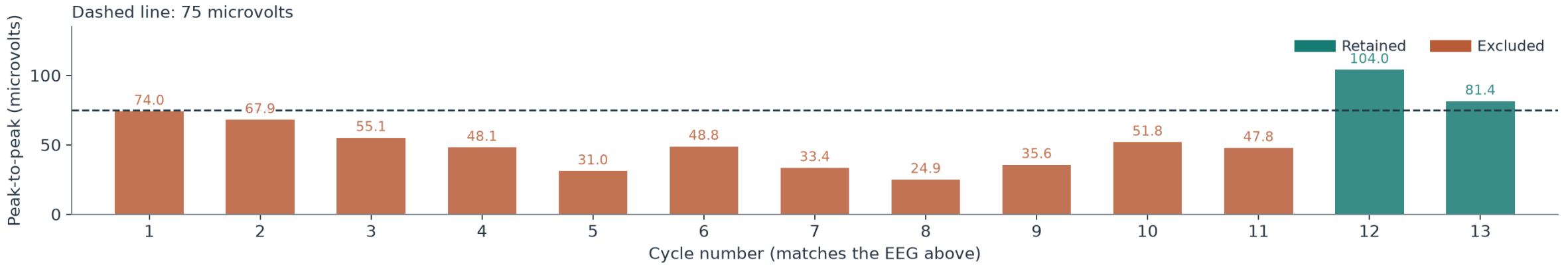
Previously liked: 5, 6, 11, 12, 13, 14, 15. Previously disliked: 1, 2, 3, 7, 8.

Which cycles does 75 microvolts retain?

NESI-NEW-01 | NESI GRDA >=0.95 | Fz-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





Retained: 2 of 13 previously accepted cycles.

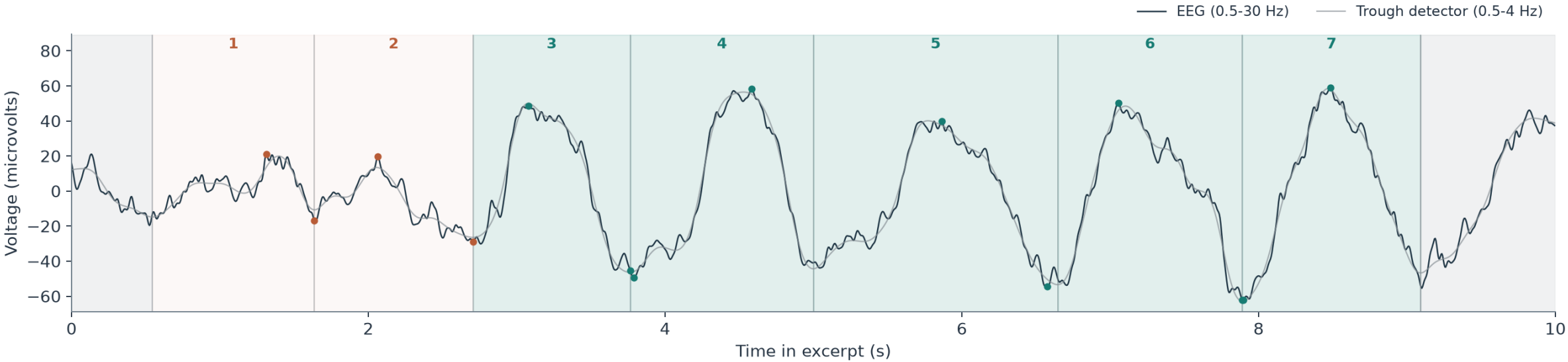
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

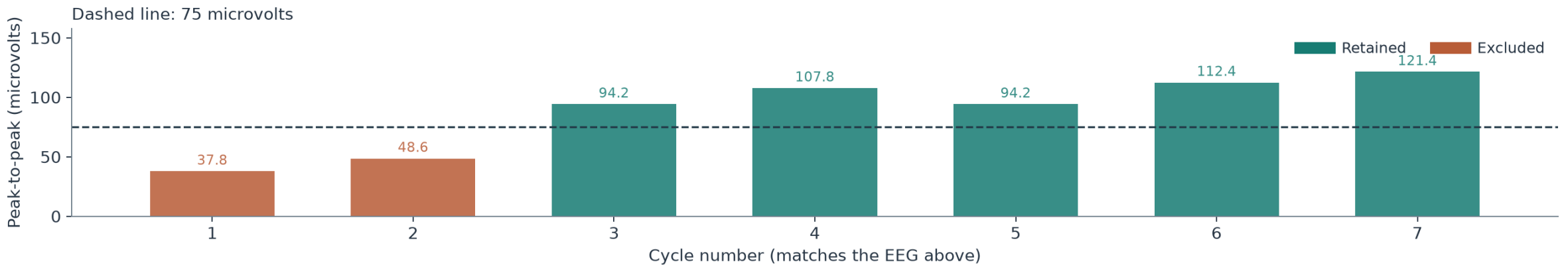
NESI microvolt scale follows the published export convention; source MAT unit labels are absent.

Which cycles does 75 microvolts retain?

NESI-NEW-02 | NESI GRDA >=0.95 | Fp1-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





Retained: 5 of 7 previously accepted cycles.

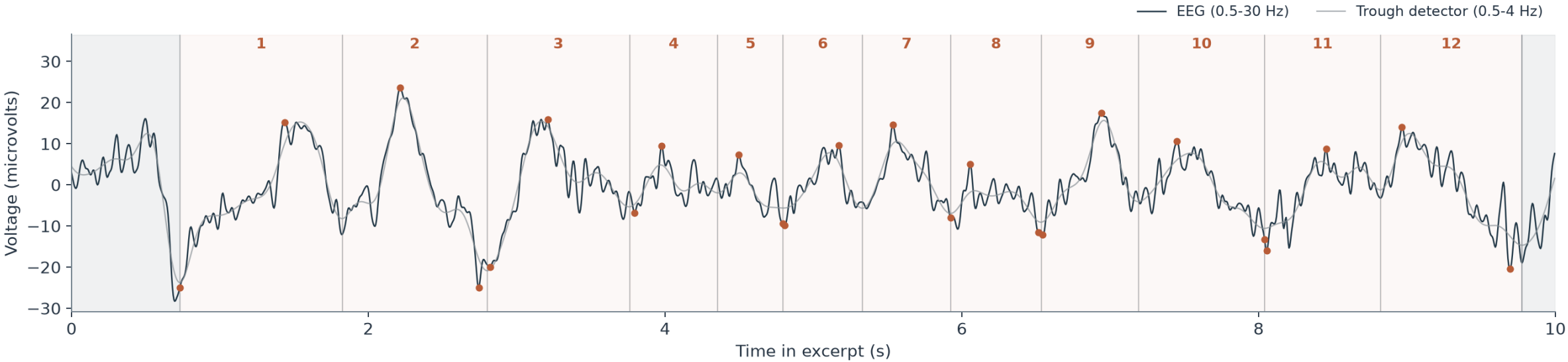
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

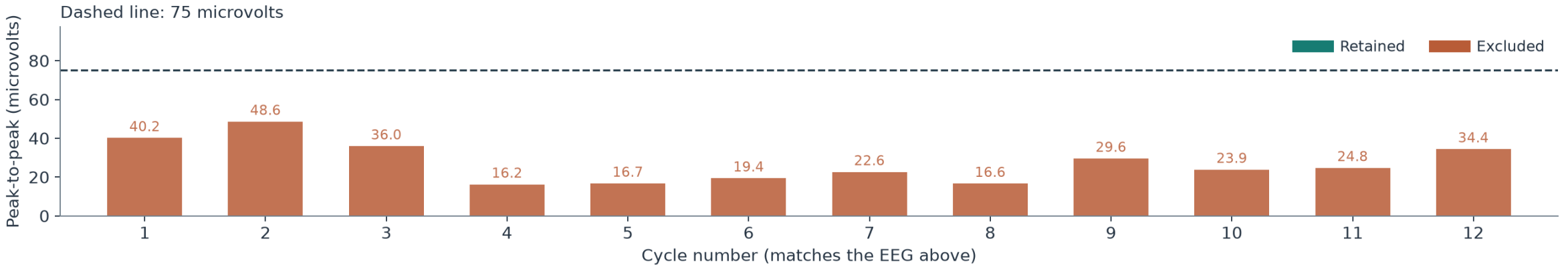
NESI microvolt scale follows the published export convention; source MAT unit labels are absent.

Which cycles does 75 microvolts retain?

NESI-NEW-03 | NESI GRDA >=0.95 | F7-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





Retained: 0 of 12 previously accepted cycles.

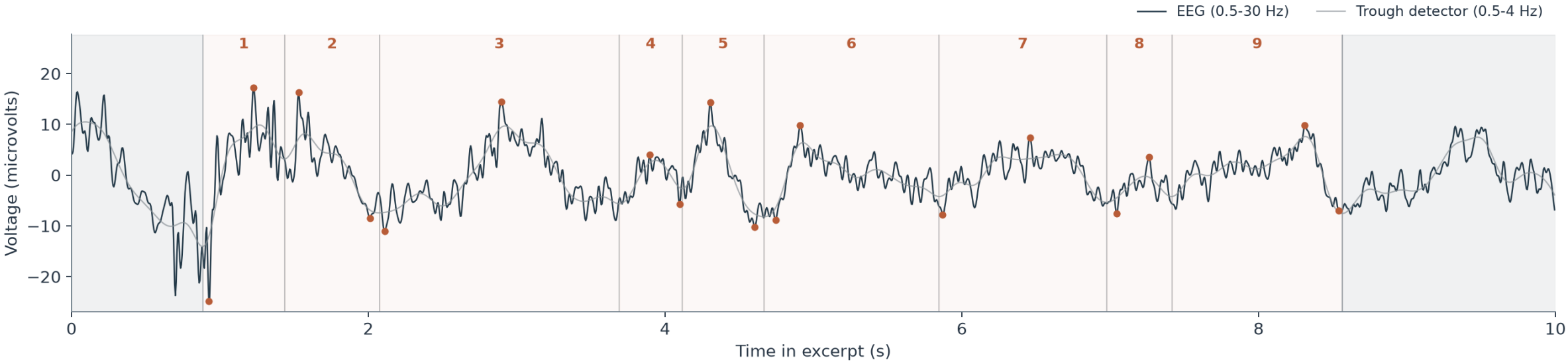
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

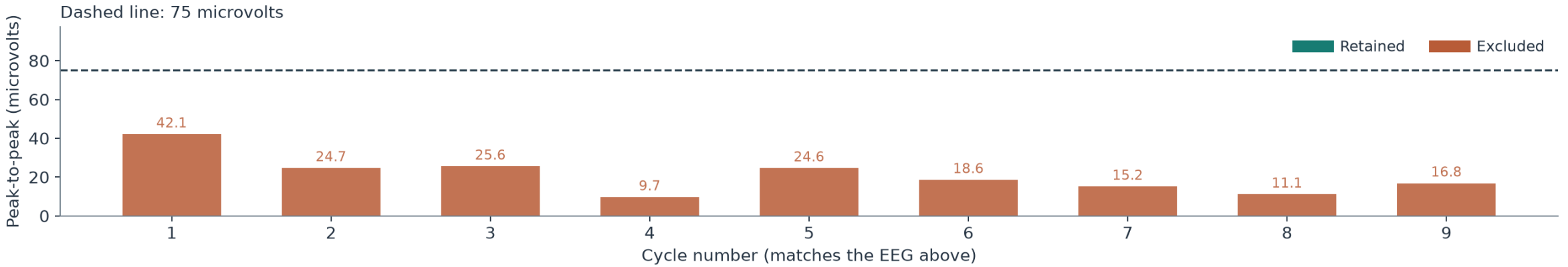
NESI microvolt scale follows the published export convention; source MAT unit labels are absent.

Which cycles does 75 microvolts retain?

NESI-NEW-04 | NESI GRDA >=0.95 | C3-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





Retained: 0 of 9 previously accepted cycles.

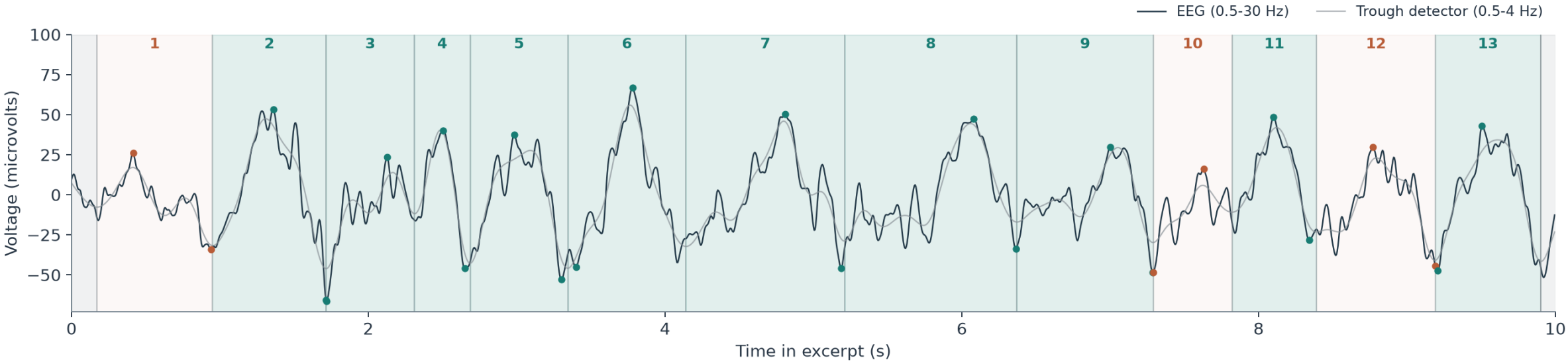
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

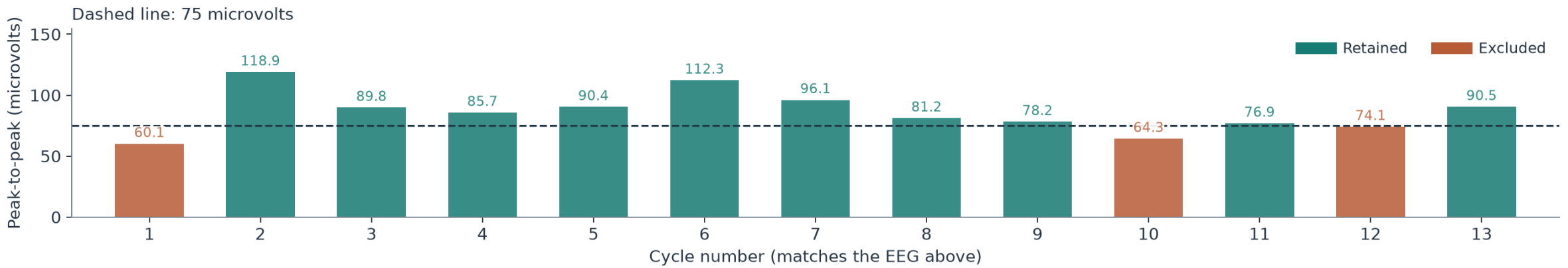
NESI microvolt scale follows the published export convention; source MAT unit labels are absent.

Which cycles does 75 microvolts retain?

MASS-NEW-01 | MASS N3 | Fz-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





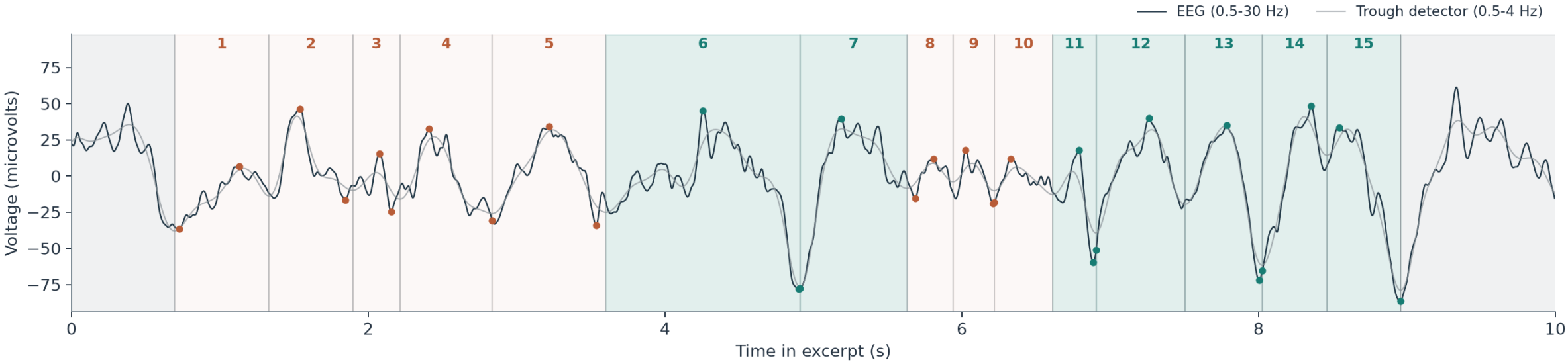
Retained: 10 of 13 previously accepted cycles.

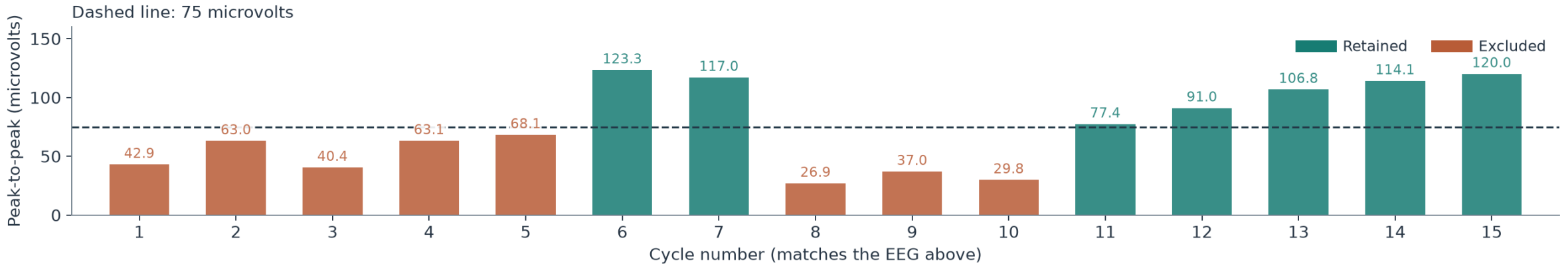
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

Which cycles does 75 microvolts retain?

MASS-NEW-02 | MASS N3 | Fp1-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





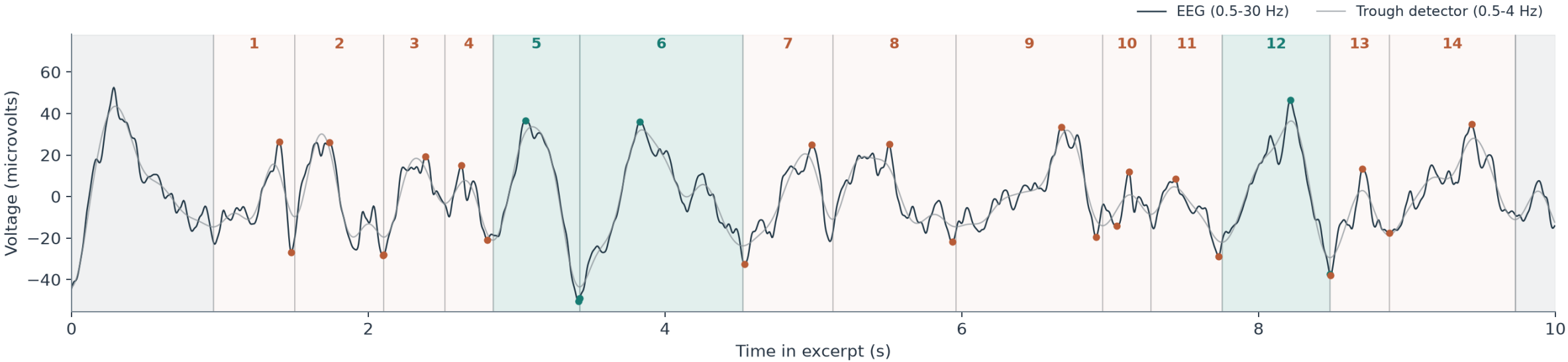
Retained: 7 of 15 previously accepted cycles.

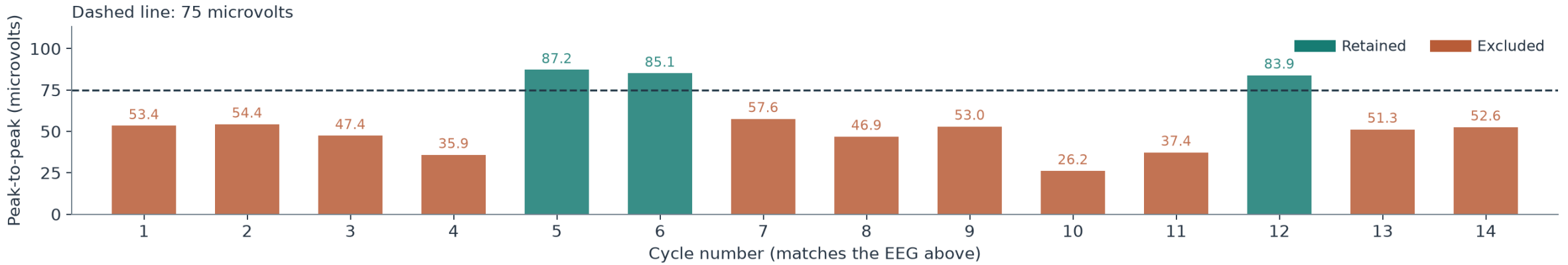
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

Which cycles does 75 microvolts retain?

MASS-NEW-03 | MASS N3 | F7-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





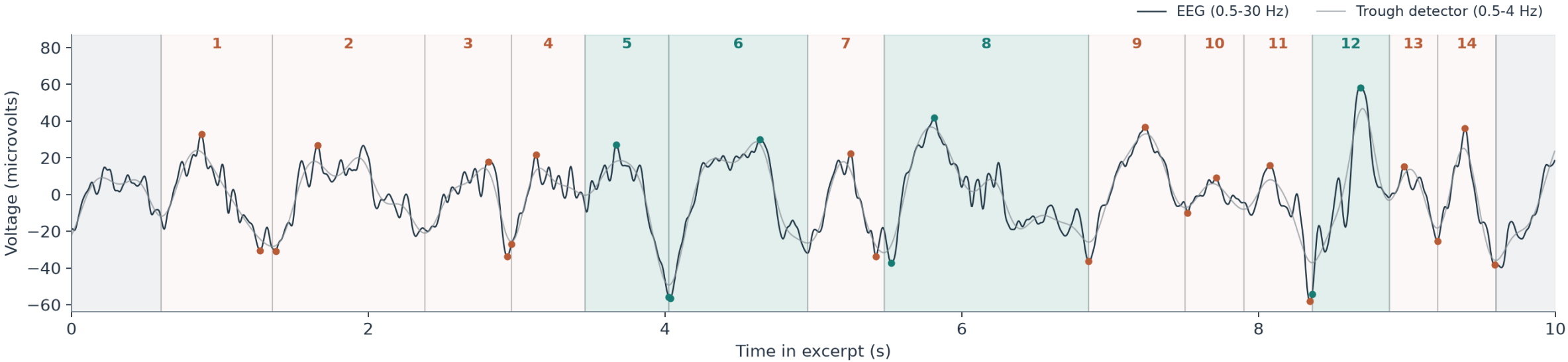
Retained: 3 of 14 previously accepted cycles.

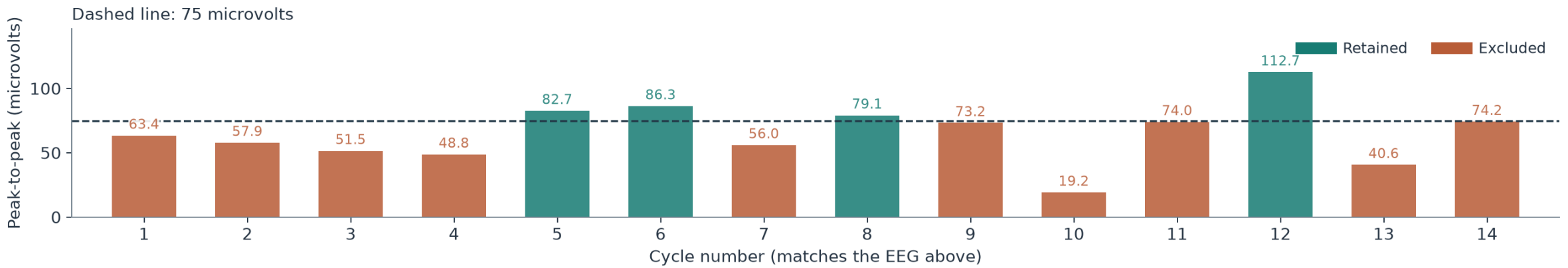
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

Which cycles does 75 microvolts retain?

MASS-NEW-04 | MASS N3 | C3-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





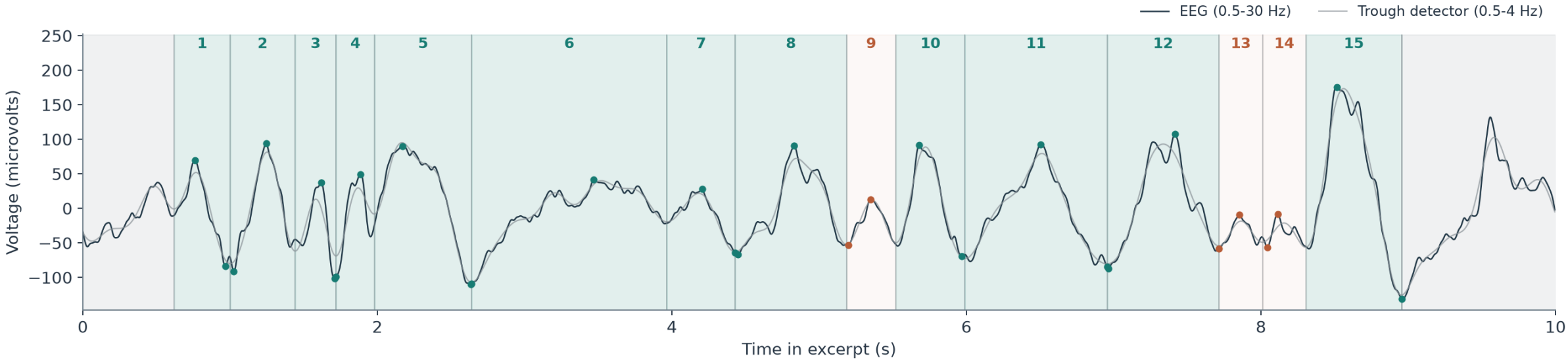
Retained: 4 of 14 previously accepted cycles.

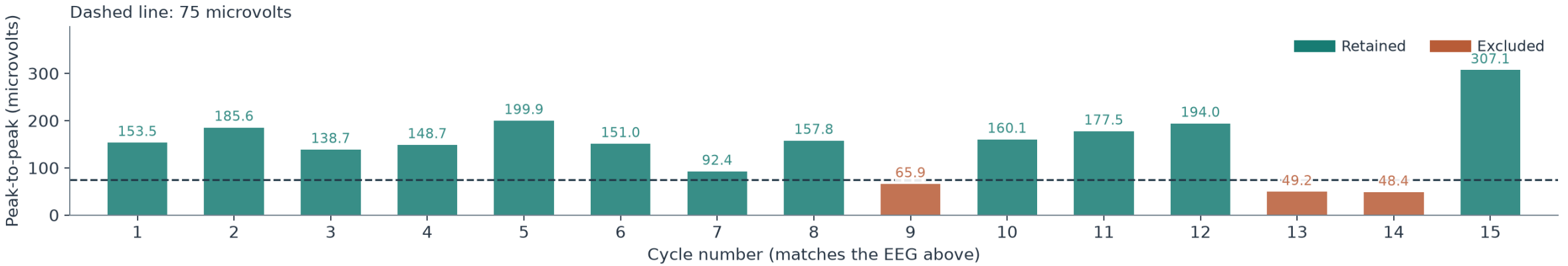
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

Which cycles does 75 microvolts retain?

LENS-NEW-01 | LENS N3 | Fz-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





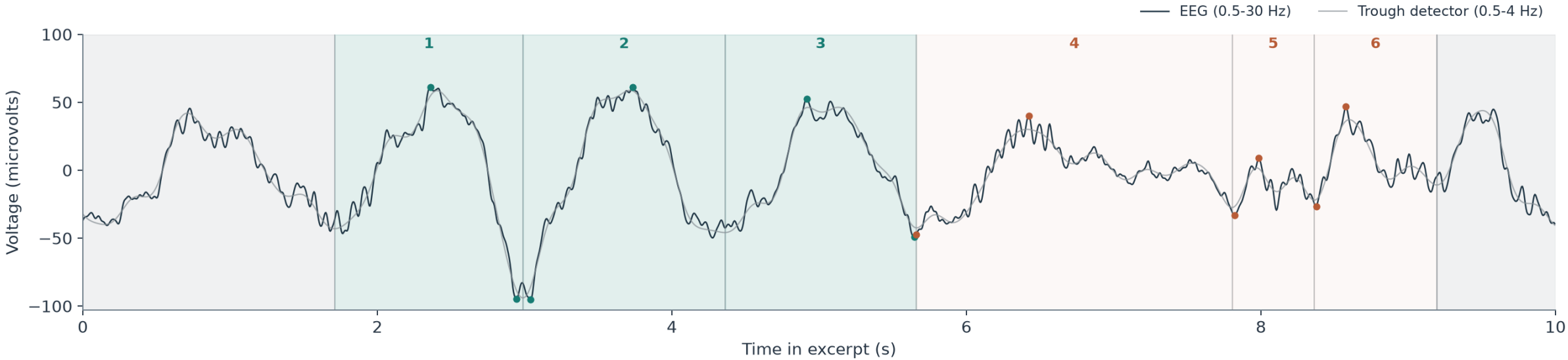
Retained: 12 of 15 previously accepted cycles.

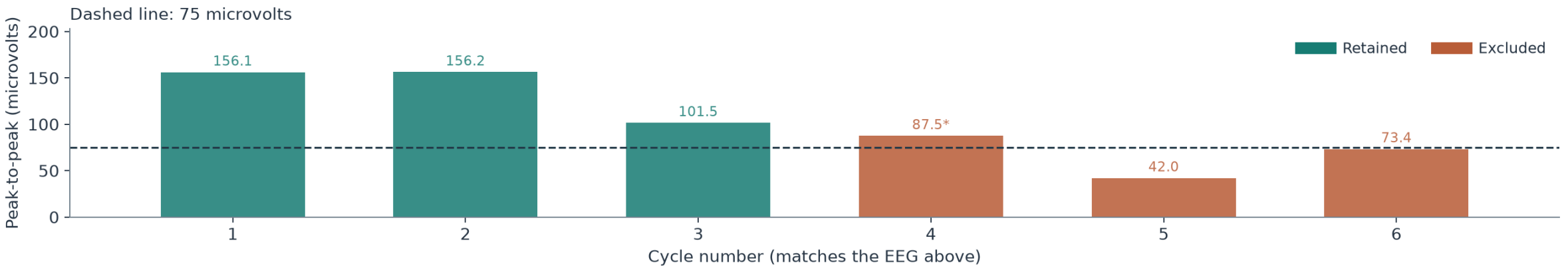
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

Which cycles does 75 microvolts retain?

LENS-NEW-02 | LENS N3 | Fp1-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





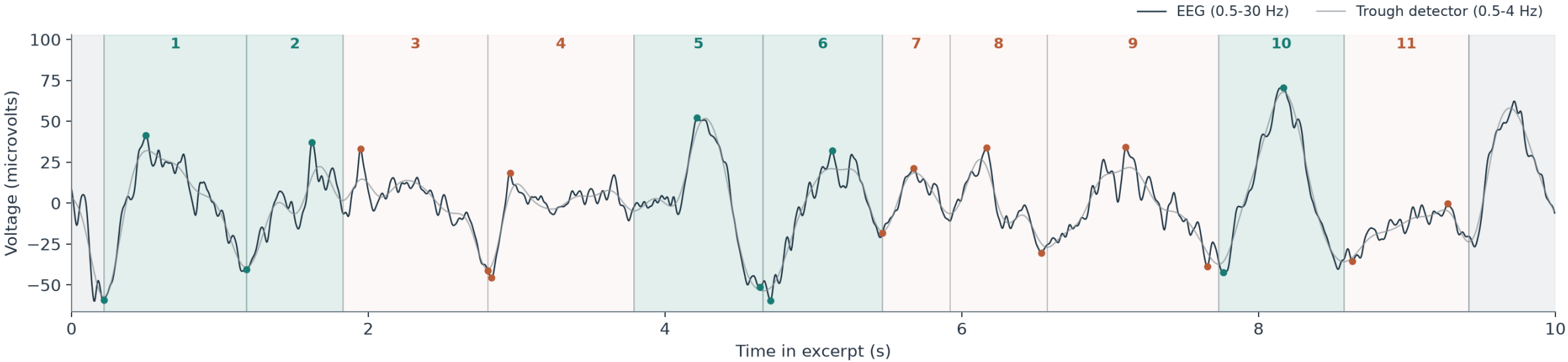
Retained: 3 of 5 previously accepted cycles.

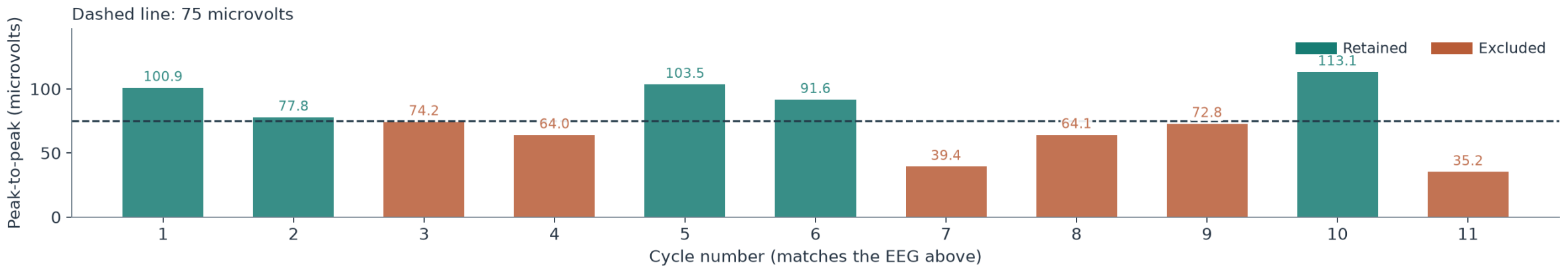
* Already excluded by the duration or constant-wave rule, even if amplitude reaches 75.

Which cycles does 75 microvolts retain?

LENS-NEW-03 | LENS N3 | F7-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





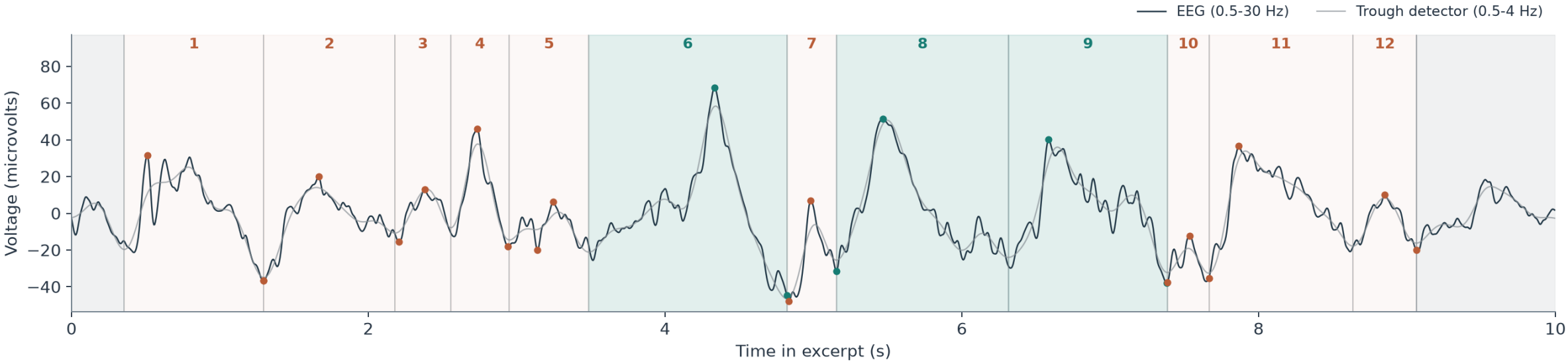
Retained: 5 of 11 previously accepted cycles.

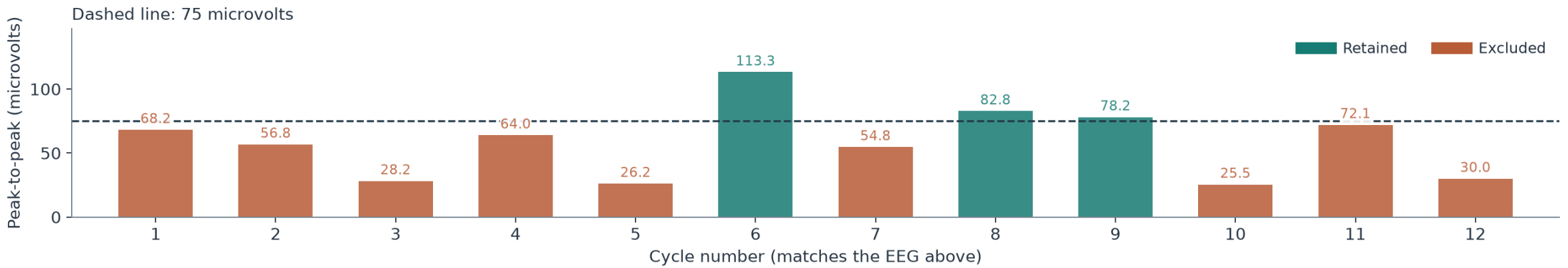
Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.

Which cycles does 75 microvolts retain?

LENS-NEW-04 | LENS N3 | C3-CAR | Fresh example selected without amplitude or cycle-result screening

Review: any retained waves that look borderline, or excluded waves that look clearly formed?





Retained: 3 of 12 previously accepted cycles.

Dots mark the maximum and minimum used for each amplitude. Gray end regions are incomplete waves.